ORGANIC MOLECULES

Module map

- Module 03: Organic Molecules
- Connected lab: lab-03_microscopy.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 03

Organic Molecules

1 Carbon skeletons

2 Building and breaking polymers

3 Carbohydrates and lipids

4 Proteins and shape

5 Nucleic acids and information

Lab evidence

lab-03_microscopy.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

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Learning objectives

- Explain why carbon is central to biological molecule diversity.
- Distinguish monomers, polymers, dehydration synthesis, and hydrolysis.
- Compare the functions of carbohydrates and lipids.
- Connect protein shape to protein function and denaturation.
- Describe how nucleic acids encode biological information.

OBJECTIVE 1
Explain why carbon is central to biological molecule diversity.

OBJECTIVE 2
Distinguish monomers, polymers, dehydration synthesis, and hydrolysis.

OBJECTIVE 3
Compare the functions of carbohydrates and lipids.

OBJECTIVE 4
Connect protein shape to protein function and denaturation.

OBJECTIVE 5
Describe how nucleic acids encode biological information.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

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Topic sequence

- Carbon skeletons: Carbon forms diverse skeletons that make biological molecules possible.
- Building and breaking polymers: Dehydration synthesis builds polymers, and hydrolysis breaks them down.
- Carbohydrates and lipids: Carbohydrates and lipids store energy and provide structure in different ways.
- Proteins and shape: Protein function depends on amino acid sequence and three-dimensional shape.
- Nucleic acids and information: DNA and RNA store and move information using nucleotide sequences.

01
Carbon skeletons
Carbon forms diverse skeletons that make biological molecules possible.

02
Building and breaking polymers
Dehydration synthesis builds polymers, and hydrolysis breaks them down.

03
Carbohydrates and lipids
Carbohydrates and lipids store energy and provide structure in different ways.

04
Proteins and shape
Protein function depends on amino acid sequence and three-dimensional shape.

05
Nucleic acids and information
DNA and RNA store and move information using nucleotide sequences.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

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Module 03: Organic Molecules Evidence Map

- Module focus: Organic Molecules
- Central claim: Organic Molecules explains carbon chemistry with polymer assembly and biological molecule evidence.
- Key nodes to connect: Carbon skeletons, Building and breaking polymers, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-03_microscopy.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

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Module 03: Organic Molecules Reasoning Sequence

- Module focus: Organic Molecules
- Trace the model: Carbon skeletons -> Building and breaking polymers -> Carbohydrates and lipids -> Proteins and shape
- Inputs become outputs: Carbon skeletons, Building and breaking polymers -> Explain why carbon is central to biological molecule diversity., Distinguish monomers, polymers, dehydration synthesis, and hydrolysis.
- Feedback check: lab-03_microscopy.md evidence checks whether students can explain carbohydrates and lipids.

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

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Terms as evidence handles

- Organic molecule: Carbon-based molecule associated with living systems.
- Monomer: Small building block that can join into a polymer.
- Polymer: Large molecule built from repeating subunits.
- Carbohydrate: Sugar or sugar polymer used for energy or structure.
- Lipid: Hydrophobic molecule including fats, phospholipids, and steroids.
- Protein: Polymer of amino acids that performs cellular work.

ORGANIC MOLECULE
Carbon-based molecule associated with living systems.

MONOMER
Small building block that can join into a polymer.

POLYMER
Large molecule built from repeating subunits.

CARBOHYDRATE
Sugar or sugar polymer used for energy or structure.

LIPID
Hydrophobic molecule including fats, phospholipids, and steroids.

PROTEIN
Polymer of amino acids that performs cellular work.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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Lab connection

- Lab file: lab-03_microscopy.md
- Lab evidence should test or illustrate: Carbohydrates and lipids
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Carbon skeletons

EVIDENCE
lab-03_microscopy.md

CLAIM
Explain why carbon is central to biological molecule diversity.

Teaching note: Make the lab connection explicit before students encounter the activity.

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Contrast check

- Do not stop at: Carbon forms diverse skeletons that make biological molecules possible.
- Stronger explanation: DNA and RNA store and move information using nucleotide sequences.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Carbon forms diverse skeletons that make biological molecules possible.

DEEPER
DNA and RNA store and move information using nucleotide sequences.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

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Module 03: Organic Molecules Retrieval and Lab Check

- Module focus: Organic Molecules
- Retrieval prompt: Why can carbon make many different biological molecules?
- Answer check: Explain why carbon is central to biological molecule diversity.
- Required terms: Organic molecule, Monomer, Polymer
- Lab connection: lab-03_microscopy.md; lab-03_microscopy.md supplies evidence for microscope observations that connect scale, structure, and biological function.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. Why is carbon the backbone of most biological molecules?
- 2. Cells join two monomers into a polymer by removing a water molecule. This reaction is called:
- 3. A protein is heated until it loses its three-dimensional shape and stops working. This change is called:
- 4. Which pairing of molecule and job is correct?

QUESTION 1

Key: C

Why is carbon the backbone of most biological molecules?

QUESTION 2

Key: A

Cells join two monomers into a polymer by removing a water molecule. This reaction is called:

QUESTION 3

Key: D

A protein is heated until it loses its three-dimensional shape and stops working. This change is called:

QUESTION 4

Key: B

Which pairing of molecule and job is correct?

Teaching note: Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

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Synthesis and exit ticket

- Exit claim: explain carbon skeletons using evidence from lab-03_microscopy.md.
- Must include: Organic molecule, Monomer, Polymer.
- Revision prompt: what evidence would change your explanation?

CLAIM
Carbon skeletons

EVIDENCE
lab-03_microscopy.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.